Tool 1 — Neurodata Governance Decision Tool

Supplementary Table 6

Scientist-facing decision tool for assigning a provisional functional-exposure tier (5, 4, 3, 2, 1) to the particular neurodata object or artefact under assessment and identifying the corresponding baseline safeguards. The assessment considers technical form together with identifiability, traceability, inferential capacity, reconstruction potential and computational persistence. Model incorporation triggers further assessment but does not automatically determine Tier 5. A tier is confirmed only when all applicable steps and required reviews are complete; an Unknown answer produces a provisional tier; a Tool 3 referral identifies a candidate tier.

Step	Question	Answer	Risk tier	Required action
Applicability checkpoint	Which statutory access regimes and ex ante data-protection, privacy-impact or privacy-risk assessment requirements may apply to the proposed activity? Consider the data category, holder, purpose, jurisdiction and application date, including EHDS where relevant. Assess legal applicability separately from functional-exposure tier.	Complete checkpoint	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record potentially applicable regimes and assessment requirements in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Record any unresolved applicability question and refer it to the responsible legal or institutional reviewer in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Continue to Step 1. Completion of this checkpoint does not resolve or waive an outstanding legal requirement.
1	Does this object contain, represent or derive from human neurodata? Examples include MRI, fMRI, EEG, MEG, ECoG, neural implant data, BCI recordings, neural features, embeddings, predictions, or AI models trained using neurodata.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record why the object falls outside this Tool's scope in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Record the outcome as "Not applicable" in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Follow the institution's ordinary data-governance procedures. End this Tool.
1	Does this object contain, represent or derive from human neurodata? Examples include MRI, fMRI, EEG, MEG, ECoG, neural implant data, BCI recordings, neural features, embeddings, predictions, or AI models trained using neurodata.	Yes	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the object's human-neurodata source or relationship in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Continue to Step 2A.

Step	Question	Answer	Risk tier	Required action
1	Does this object contain, represent or derive from human neurodata? Examples include MRI, fMRI, EEG, MEG, ECoG, neural implant data, BCI recordings, neural features, embeddings, predictions, or AI models trained using neurodata.	Unsure	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the uncertainty in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Obtain clarification from the Principal Investigator, data steward or technical lead before beginning the proposed sharing or analysis. Pause this assessment. Return to Step 1 when the evidence supports “Yes” or “No”.
2A	Has this neurodata, or information derived from it, been incorporated into a trained AI model or persistent computational infrastructure? Examples include model weights, foundation models, federated-learning systems, synthetic-data generators, model gradients, or shared embedding repositories.	Yes	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the model or infrastructure and the form of incorporation in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Mark Tool 3 (Supplementary Table 8) as required. Continue to Step 2B.
2A	Has this neurodata, or information derived from it, been incorporated into a trained AI model or persistent computational infrastructure? Examples include model weights, foundation models, federated-learning systems, synthetic-data generators, model gradients, or shared embedding repositories.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the basis for this answer in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Continue to Step 3.

Step	Question	Answer	Risk tier	Required action
2A	Has this neurodata, or information derived from it, been incorporated into a trained AI model or persistent computational infrastructure? Examples include model weights, foundation models, federated-learning systems, synthetic-data generators, model gradients, or shared embedding repositories.	Unknown	Tier 5	Record provisional Tier 5 in Neurodata Governance Assessment Record (Schedule B of the MDTAA) and apply the Tier 5 safeguards specified in Annexure 1 §3.5 to the relevant object and activity. Request technical confirmation of whether incorporation has occurred. Continue to Step 3 to identify additional exposure and safeguards. When clarification is available, return to Step 2A: if "Yes", continue to Step 2B; if "No", document resolution of this provisional finding and continue to Step 3. Do not finalise a lower tier or proceed with affected sharing, transfer or deployment while this uncertainty remains unresolved. For legacy datasets with incomplete provenance, this is a precautionary provisional classification; it does not establish that model incorporation or continuing exposure occurred. Resolve it by documenting whether and how the neurodata entered a model or persistent computational infrastructure, then reassess under Steps 2A and 2B.
2B	Is there evidence, or a reasonable possibility, that neural information persists within the model or computational infrastructure in a manner capable of supporting identification, inference, reconstruction, extraction or profiling of individuals or groups? Consider model memorisation, inversion, membership inference, extraction attacks, reconstructive outputs, shared embeddings and other forms of persistent representation.	Yes	Tier 5	Record a provisional Tier 5 condition in Neurodata Governance Assessment Record (Schedule B of the MDTAA) and apply Tier 5 safeguards provisionally under Annexure 1 §3.5. Record the evidence or credible technical assessment supporting continuing functional exposure in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Initiate or complete Tool 3 (Supplementary Table 8) and apply the Tool 3 (Supplementary Table 8) return procedure below. Continue to Step 3 to identify additional exposure and safeguards. Tier 5 remains unconfirmed until the required assessment establishes its criteria.
2B	Is there evidence, or a reasonable possibility, that neural information persists within the model or computational infrastructure in a manner capable of supporting identification, inference, reconstruction, extraction or profiling of individuals or groups? Consider model memorisation, inversion, membership inference, extraction attacks, reconstructive outputs, shared embeddings and other forms of persistent representation.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the supporting technical assessment in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Complete Tool 3 (Supplementary Table 8) because Step 2A was answered "Yes"; this answer does not remove that requirement. Apply the Tool 3 (Supplementary Table 8) return procedure below. Continue to Step 3. Do not remove an existing higher-tier finding solely on the basis of this answer.

Step	Question	Answer	Risk tier	Required action
2B	Is there evidence, or a reasonable possibility, that neural information persists within the model or computational infrastructure in a manner capable of supporting identification, inference, reconstruction, extraction or profiling of individuals or groups? Consider model memorisation, inversion, membership inference, extraction attacks, reconstructive outputs, shared embeddings and other forms of persistent representation.	Unknown	Tier 5	Record provisional Tier 5 in Neurodata Governance Assessment Record (Schedule B of the MDTAA) and apply Tier 5 safeguards under Annexure 1 §3.5. Record the unresolved exposure question in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Obtain appropriate specialist technical, legal, ethics or institutional review and complete Tool 3 (Supplementary Table 8). Apply the Tool 3 (Supplementary Table 8) return procedure below. Continue to Step 3 to identify additional exposure and safeguards. Do not finalise a lower tier or proceed with affected sharing, transfer or deployment until the uncertainty is resolved.
3	Is this object a derived neurodata product rather than the original neurodata? Examples include embeddings, feature vectors, neural fingerprints, cognitive profiles, classifiers, predictions, or reconstructed neural representations.	Yes	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the source neurodata and the transformation that produced the object in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Continue to Step 4.
3	Is this object a derived neurodata product rather than the original neurodata? Examples include embeddings, feature vectors, neural fingerprints, cognitive profiles, classifiers, predictions, or reconstructed neural representations.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the basis for treating the object as original neurodata in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Continue to Step 5 to assess direct identification and traceability.
3	Is this object a derived neurodata product rather than the original neurodata? Examples include embeddings, feature vectors, neural fingerprints, cognitive profiles, classifiers, predictions, or reconstructed neural representations.	Unknown	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Obtain provenance or processing information from the data steward or technical lead. Retain all existing tier findings and safeguards. Pause progression beyond this step. Return to Step 3 when the evidence supports “Yes” or “No”.

Step	Question	Answer	Risk tier	Required action
4	Can this derived object reasonably identify or trace an individual, or reveal, predict, reconstruct or profile neurological, cognitive, emotional, or behavioural states, traits or status concerning an individual or sufficiently defined group, including consideration of additional information reasonably available to the intended controlled-access recipient or, where public release is proposed, in the reasonably foreseeable public-access environment? Apply Annexure 1 §4.1.6 and the distinction in §2.1 between ordinary aggregate description and material group-level Functional Exposure.	Yes	Tier 4	Record a Tier 4 finding and the relevant capability in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Restrict access to controlled access; document permitted uses and prohibited inferences in Neurodata Governance Assessment Record (Schedule B of the MDTAA); prohibit unauthorised profiling, reconstruction and general-purpose AI training; and maintain provenance records. Obtain specialist governance review and complete Tool 3 (Supplementary Table 8). Complete Neuro-TADT (Tool 2) (Supplementary Table 7) before external sharing or international transfer. Retain any higher tier finding. Continue to Step 5.
4	Can this derived object reasonably identify or trace an individual, or reveal, predict, reconstruct or profile neurological, cognitive, emotional, or behavioural states, traits or status concerning an individual or sufficiently defined group, including consideration of additional information reasonably available to the intended controlled-access recipient or, where public release is proposed, in the reasonably foreseeable public-access environment? Apply Annexure 1 §4.1.6 and the distinction in §2.1 between ordinary aggregate description and material group-level Functional Exposure.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the evidence supporting this answer in Neurodata Governance Assessment Record (Schedule B of the MDTAA), including consideration of additional information reasonably available to the intended recipient. Continue to Step 5.
4	Can this derived object reasonably identify or trace an individual, or reveal, predict, reconstruct or profile neurological, cognitive, emotional, or behavioural states, traits or status concerning an individual or sufficiently defined group, including consideration of additional information reasonably available to the intended controlled-access recipient or, where public release is proposed, in the reasonably foreseeable public-access environment? Apply Annexure 1 §4.1.6 and the distinction in §2.1 between ordinary aggregate description and material group-level Functional Exposure.	Unknown	Tier 4	Record provisional Tier 4 and the capability requiring assessment in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Apply the safeguards specified in the Step 4 “Yes” row provisionally. Complete Tool 3 (Supplementary Table 8) and obtain the specialist review needed to resolve the uncertainty. Retain any higher tier finding. Continue to Step 5.
5	Does the object contain direct identifiers, recognisable identifying anatomy, identifying acquisition metadata, or intrinsic information that reasonably permits participant identification or tracing beyond linkage through a separately protected code or mapping? Examples include names, identifying DICOM headers, non-defaced facial anatomy, identifying voice or video, or device identifiers and timestamps that enable tracing. Being a raw EEG, MEG, ECoG or MRI acquisition is not, by itself, a “Yes”; linkage solely through a protected code or mapping is assessed at Step 6.	Yes	Tier 3	Record a Tier 3 finding and the identifiers or intrinsic traceability identified in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Store the object in a secure controlled environment with encryption, role-based access, audit logs, and formal access approval. Remove unnecessary identifiers where feasible and restrict downloads and onward sharing. Complete Neuro-TADT (Tool 2) (Supplementary Table 7) before international transfer or external access. Retain any higher tier finding. Continue to Step 6.

Step	Question	Answer	Risk tier	Required action
5	Does the object contain direct identifiers, recognisable identifying anatomy, identifying acquisition metadata, or intrinsic information that reasonably permits participant identification or tracing beyond linkage through a separately protected code or mapping? Examples include names, identifying DICOM headers, non-defaced facial anatomy, identifying voice or video, or device identifiers and timestamps that enable tracing. Being a raw EEG, MEG, ECoG or MRI acquisition is not, by itself, a “Yes”; linkage solely through a protected code or mapping is assessed at Step 6.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the basis for concluding that the specified identifiers and intrinsic traceability are absent in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Continue to Step 6.
5	Does the object contain direct identifiers, recognisable identifying anatomy, identifying acquisition metadata, or intrinsic information that reasonably permits participant identification or tracing beyond linkage through a separately protected code or mapping? Examples include names, identifying DICOM headers, non-defaced facial anatomy, identifying voice or video, or device identifiers and timestamps that enable tracing. Being a raw EEG, MEG, ECoG or MRI acquisition is not, by itself, a “Yes”; linkage solely through a protected code or mapping is assessed at Step 6.	Unknown	Tier 3	Record provisional Tier 3 in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Apply the safeguards specified in the Step 5 “Yes” row provisionally. Assign an identifier and traceability review to the responsible data steward or technical reviewer. Retain any higher tier finding. Continue to Step 6. Return to Step 5 when the review supports “Yes” or “No”.
6	Does the object still contain participant-level neurodata that can be linked back to an individual through a code, repository record, or institutional key? Example: A pseudonymised EEG dataset linked to a participant identifier held by the originating institution, or a de-identified MRI scan that can be re-linked through a secure repository key.	Yes	Tier 2	Record a Tier 2 finding, the linkage mechanism and the responsible custodian in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Use controlled-access repositories; separate and protect the linkage key; encrypt data in storage and transit; maintain access logs; and restrict re-identification and onward sharing. Put any required Data Use or Transfer Agreement in place before the relevant access or transfer. Where another institution holds the key, record its custody and restrictions without obtaining or disclosing the key. Retain any higher tier finding. Proceed to the Final Decision.
6	Does the object still contain participant-level neurodata that can be linked back to an individual through a code, repository record, or institutional key? Example: A pseudonymised EEG dataset linked to a participant identifier held by the originating institution, or a de-identified MRI scan that can be re-linked through a secure repository key.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the evidence supporting the absence of linkage through a code, repository record, or institutional key in Neurodata Governance Assessment Record (Schedule B of the MDTAA). If a candidate, provisional, or confirmed Tier 3, Tier 4 or Tier 5 finding remains active, proceed to the Final Decision. Otherwise, continue to Step 7.

Step	Question	Answer	Risk tier	Required action
6	Does the object still contain participant-level neurodata that can be linked back to an individual through a code, repository record, or institutional key? Example: A pseudonymised EEG dataset linked to a participant identifier held by the originating institution, or a de-identified MRI scan that can be re-linked through a secure repository key.	Unknown	Tier 2	Record provisional Tier 2 in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Apply the linkage safeguards specified in the Step 6 “Yes” row to any existing or potentially retained linkage mechanism. Request confirmation from the data holder or repository custodian about whether participant-level linkage remains possible. Retain any higher tier finding. Proceed to the Final Decision with the uncertainty unresolved. When confirmation is available, return to Step 6 and follow the “Yes” or “No” branch.
7	Does the object contain only aggregated or summary-level information with no participant-level records or persistent participant trace? Example: A published study reporting only group-average brain activation patterns, summary statistics, or aggregate connectivity measures with no participant-level records or re-identifiable individual data.	Yes	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record the aggregation or summarisation method and the evidence that no participant-level records or persistent participant trace remain in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Continue to Step 8.
7	Does the object contain only aggregated or summary-level information with no participant-level records or persistent participant trace? Example: A published study reporting only group-average brain activation patterns, summary statistics, or aggregate connectivity measures with no participant-level records or re-identifiable individual data.	No	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record “Reassessment required” in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Review the object’s technical form, provenance and exposure characteristics, then revisit Steps 3–6 and correct any answers affected by the review. If the evidence supports a tier, record the assigned tier and continue the classification workflow. For participant-level data that are de-identified and have no reasonably available linkage mechanism, assign Tier 2A; if a linkage key or other reasonably available linkage mechanism is retained, consider Tier 2B. If the data remain identifiable or intrinsically traceable, assign the applicable higher tier. Refer for specialist review only where the available evidence is insufficient or the appropriate tier remains uncertain. Resume at the relevant step when the review provides sufficient evidence.
7	Does the object contain only aggregated or summary-level information with no participant-level records or persistent participant trace? Example: A published study reporting only group-average brain activation patterns, summary statistics, or aggregate connectivity measures with no participant-level records or re-identifiable individual data.	Unknown	No tier is assigned at this step. Follow the instructions in the Required action column. Any tier identified at an earlier step remains applicable unless resolved through documented reassessment.	Record that aggregate-only status has not been established in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Request technical review of participant-level records, persistent traces and the aggregation or summarisation method. Pause classification and retain any existing safeguards. Do not approve release through this Tool while the uncertainty remains unresolved. Return to Step 7 when the evidence supports “Yes” or “No”.

Step	Question	Answer	Risk tier	Required action
8	Could an individual or small group still reasonably be identified through rare conditions, small sample sizes, unique characteristics, or accompanying metadata? Example: A dataset describing a single participant with a rare neurological disorder, a very small cohort from one clinic, or records containing unique demographic or geographic metadata that could enable re-identification.	Yes	Tier 2	Record Tier 2 as the minimum applicable tier and identify the disclosure mechanism in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Increase aggregation or suppress disclosive information where feasible. Determine and record the necessary access restrictions, including controlled access where required. Reassess the modified object before release, repeating Steps 7–8 and any earlier step affected by the modification. Retain any higher tier finding unless documented reassessment resolves it. Proceed to the Final Decision.
8	Could an individual or small group still reasonably be identified through rare conditions, small sample sizes, unique characteristics, or accompanying metadata? Example: A dataset describing a single participant with a rare neurological disorder, a very small cohort from one clinic, or records containing unique demographic or geographic metadata that could enable re-identification.	No	Tier 1 Note: A Tier 1 outcome does not automatically exclude Annexure 3 in the MDTAA. Apply the exception and triggers in Annexure 1 §7.2 in the MDTAA to the particular object and proposed activity, and record the reason for completing or omitting Annexure 3 in the MDTAA. A Tier 1 outcome shall not override a higher candidate, provisional or confirmed tier identified elsewhere in the assessment.	Record a Tier 1 finding and the supporting disclosure-risk assessment in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Document the aggregation or anonymisation method and confirm that no participant-level record, persistent participant trace or linkage mechanism remains in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Check the triggers and exceptions in Annexure 1 §7.2 and record why Tool 3 (Supplementary Table 8) is required or may be omitted. Do not use this finding to override a higher candidate, provisional or confirmed tier. Proceed to the Final Decision.
8	Could an individual or small group still reasonably be identified through rare conditions, small sample sizes, unique characteristics, or accompanying metadata? Example: A dataset describing a single participant with a rare neurological disorder, a very small cohort from one clinic, or records containing unique demographic or geographic metadata that could enable re-identification.	Unknown	Tier 2	Record provisional Tier 2 in Neurodata Governance Assessment Record (Schedule B of the MDTAA). Obtain a disclosure-risk review addressing rare conditions, small samples, unique characteristics, and accompanying metadata. Withhold release pending resolution of this uncertainty. Retain any higher tier finding. Proceed to the Final Decision with the uncertainty unresolved. Return to Step 8 when the review supports “Yes” or “No”.

Step	Question	Answer	Risk tier	Required action
Final Decision	Have all applicable steps and required classification reviews been completed?	Complete assessment	-	Record the highest applicable tier and whether it is provisional or confirmed. Do not finalise a lower tier while a higher candidate or provisional finding remains unresolved. Retain all safeguards whose underlying conditions remain present. Save the outcome, supporting evidence, outstanding actions and responsible reviewer in the assessment record; incorporate or reference it in Neurodata Governance Assessment Record (Schedule B of the MDTAA) where applicable. Complete or update Tools 2 (Supplementary Table 7) and 3 (Supplementary Table 8) where required, then proceed to Tool 4 (Supplementary Table 9) where contractual implementation is needed. Do not begin the proposed activity until required safeguards and approvals are in place. If public release is proposed, continue to Tool 2 / Table 7 using the Public-release pathway.